

GAME DESIGNER & TECHNOLOGIST

ASHTON

ZHANG

(SHUHAN)

SCROLL TO FOLLOW CLICK TO INTERACT



PORTFOLIO WEBSITE

shuhan0001.github.io/portfolio/

GITHUB

github.com/Shuhan0001



COGNI-STREAM

CASE STUDY / 00

Cognitive-as-a-Service

A speculative critical-design report imagining EnGram+, an invasive brain-computer interface that turns knowledge into a subscription tier and turns class into a light on your temple.

Researcher	Prototype	Module	Programme
Shuhan Zhang	Cogni-Stream / EnGram+	Speculative Design · TFT00068H / TFT00125M	University of York



WATCH

Test video

youtu.be/ltBEEefctw



EXPLORE

Project GitHub

github.com/Shuhan0001/Cogni-Stream_Website

CONCEPT / 01

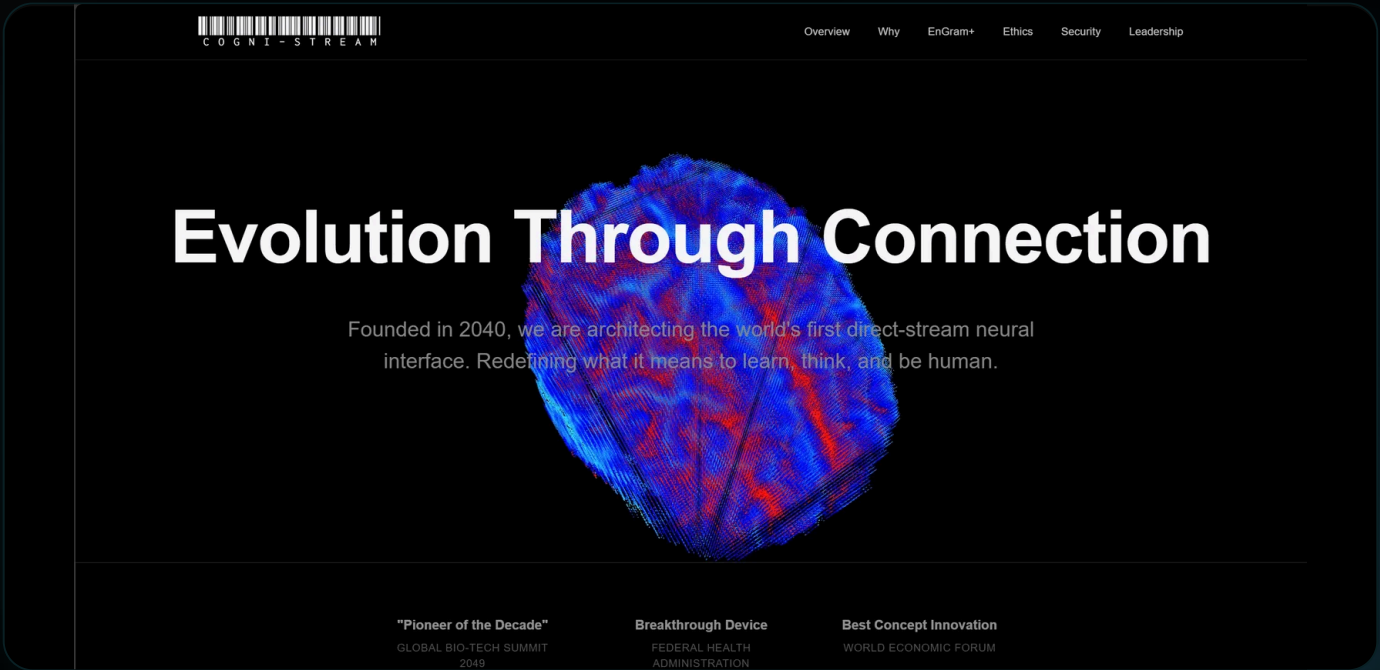
The Premise

Founded in 2040, Cogni-Stream is a fictional techno-medical enterprise built around EnGram+, an invasive BCI device that downloads knowledge directly into the brain.

Access is not earned through study. It is purchased through a subscription tier, and the price paid determines how much of a person's own mind they get to keep.

CRITICAL PROPOSITION

What happens when cognition becomes a renewable service rather than a human right?



Homepage hero - premium consumer-tech language used to make cognitive extraction feel desirable.

CONCEPT / 02

Access becomes hierarchy

The pricing system translates economic class into cognitive bandwidth. Each card is both a product tier and a social position.

Public \$0 / lifetime	INCLUDED STREAMS Global Literacy Fundamental Arithmetic	PROTOCOL Ad-supported 2KB/s bandwidth	Essential \$999 / yr	INCLUDED STREAMS General History Basic Sciences	PROTOCOL Clean Stream 50KB/s bandwidth
Professional \$9,999 / yr	INCLUDED STREAMS Advanced Mathematics Full Programming Logic	PROTOCOL Deep Retention 1MB/s bandwidth	Elite Legacy Invite only	INCLUDED STREAMS Executive Strategy Fine Arts Intuition	PROTOCOL Genetic Inheritance Uncapped bandwidth

COGNI-STREAM / 03

RESEARCH / 01

Problem Space

If knowledge becomes a neural subscription, the digital divide won't close. It will simply move inside your skull.

01. Digital Divide, at Neural Resolution

Van Dijk (2019) frames the digital divide as a structural gap in access and skill, driven by income, education, geography and age rather than technology itself. As of 2022, roughly 2.7 billion people still lacked internet access, and over half the world had no high-speed broadband (Signé, 2023). That's a direct challenge to UN SDG 4's promise of inclusive, equitable education.

EnGram+ takes that same structural gap and moves it inside the body. When knowledge is downloaded rather than taught, capital determines cognitive speed directly, and the divide is no longer something a library card can close.

COGNI-STREAM / 04

RESEARCH / 02

02. Technocapitalism and the Commodified Mind

Under technocapitalism (Suarez-Villa, 2000; 2012), knowledge and innovation are steadily absorbed into corporate-controlled infrastructure, displacing public institutions as the gatekeepers of learning. Napoleoni (2024) argues this concentration of infrastructural control lets a handful of firms accumulate wealth and influence at an unprecedented scale.

Who will ultimately control access to human cognition in the neural infrastructure of the future

Framing question raised by Napoleoni's (2024) account of technocapitalism, applied to Cogni-Stream's fictional world

In Cogni-Stream's world, education has fully migrated from public good to corporate product. Cognition itself becomes a pawn of capital: not just a shift in business model, but a descent of power from institutions into the neural sphere.

COGNI-STREAM / 05

RESEARCH / 03

03. KaaS and the Status Halo

Tzuo (2018) describes a broader shift from owning products to using services; EnGram+ applies this logic to cognition itself through an annual subscription. Sheil et al. (2024) document "Roach Motel" dark patterns: easy sign-up, punishing cancellation. EnGram+ pushes the tactic further by making non-payment visible.

When a subscription lapses, the implant's Status Halo, mounted at the temple, switches to an amber glow that anyone nearby can see. Users who can't afford renewal face public stigma long before they face a locked door.

The Status Halo: four states



ACTIVE
System is fully operational.



SYNCING
Downloading knowledge packages.



LIMITED
Subscription expired. Basic cognition only.



CRITICAL
Hardware damage or illegal operation detected.

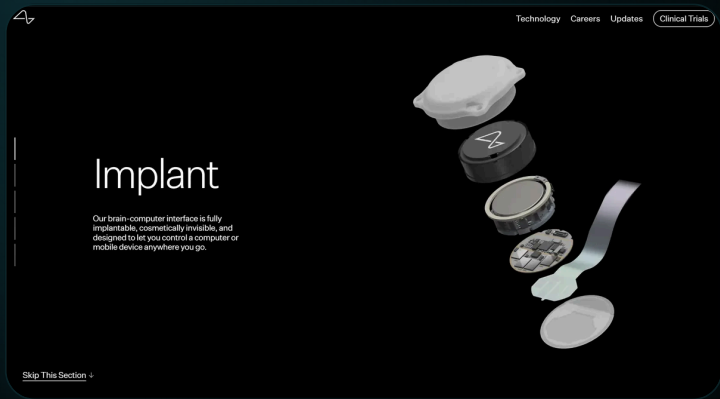
If a downgraded user can't pay for removal surgery either, they're left with a device they can't afford to keep and can't afford to take out. The lock-in is economic, social, and finally cognitive, as EnGram+ subtly reshapes how its users think, not just what they know.

COGNI-STREAM / 06

DESIGN / 01

Design Background

Three reference points shaped how Cogni-Stream looks, sounds, and lies to its users.



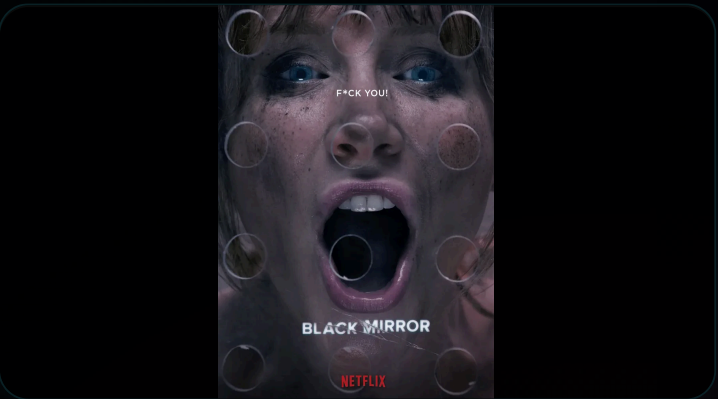
Neuralink: Clinical Aesthetics

Neuralink's real-world BCI research is the project's technical anchor. Its streamlined, consumer-electronics-white surgical robot reframes brain surgery as routine. The Cogni-Stream manual borrows this strategy directly, recasting implantation as "hardware configuration" performed by a "Certified Neural Technician."



Apple: the Visual Disguise of Power

Apple's expansive white space and precise typographic grid read as premium and transparent (Apple Inc., 2026), while quietly disguising a closed ecosystem. Cogni-Stream borrows this "Corporate Minimalism" so its predatory downgrade clauses sit inside pages elegant enough that users skim past them.



Black Mirror: Visible Class Anxiety

"Nosedive" (2016) turns social standing into a number superimposed on everyday life. It directly informed the Status Halo and the simulated X community: a device that makes economic status legible to strangers, and a feed where that visibility turns into public discrimination.

COGNI-STREAM / 07

DESIGN / 02

Inside the Feed

The simulated X community sits inside the website, modelled on the real platform's visual language. Official Cogni-Stream posts stay upbeat and evasive; a breaking-news account raises concerns about memory disruption after downgrades; and ordinary users describe what the Status Halo actually costs them in public.

Cogni-Stream @Cogni_Stream

Education inequality predates neural technology. Our mission is innovation, not policy reform.

Cogni-Stream @Cogni_Stream

EnGram+ is designed with multiple safety protocols to ensure a stable, reliable cognitive experience. In rare cases, users may notice temporary disorientation after a tier change. This is typically short-lived.

TheBreakingNews

Multiple reports suggest that when EnGram+ subscriptions lapse and users are auto-downgraded, some experience mild memory disruption. Advocacy groups are calling for greater caution around implantation.

404NotFound_Wy @Wy_2028831

My EnGram+ subscription ran out and it auto-downgraded me. Didn't think it'd be a big deal, but some of what I downloaded is just gone, and a few normal memories feel blurry too.

And this thing on my temple won't stop blinking yellow. Removal costs a surgery fee I can't afford right now. Anyone else dealt with this

● **Status Halo: Limited**

COGNI-STREAM / 08

SOURCE / OVERVIEW

From the Source

The prototype is hand-built HTML, CSS and JavaScript. These selected excerpts show how the interface turns a critical argument into a believable product system.



INCLUDED FILE
Draft wireframe

EnGram+_Web_Draft.html



LAUNCH
Live prototype

shuhan0001.github.io/Cogni-Stream

Code is not presented here as implementation detail. It is evidence of how hierarchy, trust and coercion were deliberately embedded into the interface.

01

Funnel

One path makes consent feel inevitable.

02

Status

Colour and badges establish social rank.

03

Infrastructure

The product's threat remains quietly present.

SOURCE / 01

The Funnel

The homepage offers one glowing route forward. A product-launch structure makes a predatory subscription funnel feel inevitable rather than chosen.

01

One path, framed as progress

The hierarchy moves from corporate promise to product gate without presenting a competing action. The interface is already making the ethical decision before the user reaches the pricing page.

EnGram__Web.html: corp-hero & product-gate

```
<section class="corp-hero">
  <h1>Evolution Through Connection</h1>
  <p>We are architecting the world's first
    direct-stream neural interface...</p>
</section>

<section class="product-gate">
  <h2>Meet EnGram+</h2>
  <div class="gate-card">
    <h3>Neural Port Visualization</h3>
    <div class="btn-primary">See Details</div>
  </div>
</section>
```

SOURCE / 02

Interface as Persuasion

Gold distinguishes the elite product tier before its price is understood. The colour token makes social hierarchy instantly legible.

02

Gold Is a Colour Token

The elite tier's name is gold before its price is understood. Social hierarchy is already legible in the CSS token.

EnGram__Web.html: #pricing

```
:root {
  --elite-gradient:
    linear-gradient(90deg, #d4af37, #f3e5ab);
}

.tier-elite .plan-name {
  background: var(--elite-gradient);
  -webkit-background-clip: text;
  -webkit-text-fill-color: transparent;
}
```

SOURCE / 03

Trust by class name

A single modifier class separates corporate reassurance from ordinary testimony. Visual authority is supplied before the copy is read.

03

One class decides who you trust

The verified state is structural, not editorial. The same feed component can make one voice institutional and another disposable.

EnGram__Web.html: page-screenshots

```
<div class="tweet tweet-official">
  <div class="verified-badge">
    Verified Organization
  </div>
  <p>A stable, reliable cognitive
    experience...</p>
</div>

<div class="tweet">
  My subscription ran out and it
  auto-downgraded me...
</div>
```

SOURCE / 04

Infrastructure & Ethics

A rotating point-cloud brain remains behind the homepage, turning decorative infrastructure into a constant reminder that cognition is the product.

04

A Brain Behind Everything

A rotating point-cloud brain loads behind the homepage. Decorative infrastructure becomes a constant reminder that cognition is the product.

```
initThreeJSBackground()  
  
const scene = new THREE.Scene();  
const loader = new THREE.GLTFLoader();  
  
loader.load('Model/brain_point_cloud.glb',  
function(gltf) {  
    model = gltf.scene;  
    scene.add(group);  
});  
  
function animate() {  
    requestAnimationFrame(animate);  
    model.rotation.y += 0.002;  
    renderer.render(scene, camera);  
}
```

SOURCE / 05

Ethics versus pricing

The Ethics page promises cognitive sovereignty and equitable access, while the pricing and downgrade clauses quietly negate both claims.

05

The contradiction is part of the system

The language of autonomy is rendered with the same confidence as the commercial mechanism that removes it. The tension is designed into the product, not added as commentary afterward.

```
EnGram__Web.html: page-ethics  
  
<h4>Cognitive Sovereignty</h4>  
<p>Your mind is your ultimate sanctuary.  
    We guarantee absolute user autonomy.</p>  
  
<h4>Equitable Access</h4>  
<p>Evolution should not be a privilege.  
    Our Public tier ensures essential  
    knowledge streams remain free.</p>
```

RATIONALE / 01

Design Rationale

Why multimodal

Power under technocapitalism operates across media, not within one, so a single format couldn't reveal its systemic character. Speculative design builds credible fiction as a tool for critical reflection (Dunne & Raby, 2013), which meant the project needed a website, a promotional video, and a social feed working together, not in isolation.

Why first-person

The subscription selector on the homepage invites visitors to temporarily become consumers, not observers. It's usually only at the point of picking a tier that people notice they've quietly agreed to the commodification of knowledge inside the story. That delayed self-awareness does more critical work than a lecture would (Bleecker, 2009).

Why Apple + Neuralink aesthetics

Borrowing Apple and Neuralink's high-contrast type, precise grids, and restrained palette gives the fiction enough visual credibility to be mistaken for a real product. That's the point: polished interfaces are treated here as instruments of soft power (Suarez-Villa, 2012), not neutral style. The elegance is part of the argument, not a decoration on top of it.

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REFLECTION / 01

Closing Reflection

Cogni-Stream isn't a warning about brain implants. It's a mirror held up to how technocapitalism already works.

Grounded in SDG 4's vision of equitable education and read through Van Dijk's digital divide, Suarez-Villa's technocapitalism, and Sheil et al.'s dark-pattern research, the project turns "neuro-inequality" from an abstract concept into something you can click through and feel implicated in. The right to learn, in this fiction, has quietly become an extension of purchasing power. The value of speculative design lies in expanding what we're able to imagine and question about that (Dunne & Raby, 2013), not in predicting it.

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University of York, Speculative Design (TFT00068H / TFT00125M)

AI disclosure: Google Gemini assisted with JavaScript debugging; Midjourney generated product imagery; Runway AI generated promotional video elements. All outputs were reviewed, curated and directed by the author. Full disclosure in the written report.

The Bridge of Pixel and Light

A controlled study of how lighting changes visibility, search behaviour and immersion in a 2.5D pixel environment.

Researcher

Shuhan Zhang

Prototype

Revive

Programme

BSc Interactive Media 2025 to 2026



WATCH
Gameplay video
youtu.be/-3xa17G_q5k



EXPLORE
Project GitHub
github.com/Shuhan0001/Revive

REVIVE / 01

Research Question

2.5D pixel art combines hand drawn sprites with three dimensional space. Its compressed visual language removes many of the depth cues found in conventional 3D scenes. Lighting therefore carries more responsibility for separating objects, revealing space and directing attention.

The study asks how changes in illumination affect scene readability and player behaviour. It focuses on visibility, search efficiency, attention and exploration. The central premise is that lighting is a structural cue rather than a purely aesthetic layer. The analysis is informed by visual perception research from Ware and navigation research from Darken and Sibert. Cognitive load and limited attention provide the main behavioural framework.



Fig 1. A local light source can reveal spatial information that is otherwise compressed by the pixel art style.

REVIVE / 02

Research Method

The experiment used one fixed environment with three lighting conditions. Day provided high visibility. Sunset reduced ambient illumination while retaining strong environmental structure. Night removed most ambient light and made the handheld lantern central to navigation and search. Scene layout, fragment positions, controls and mechanics remained unchanged.

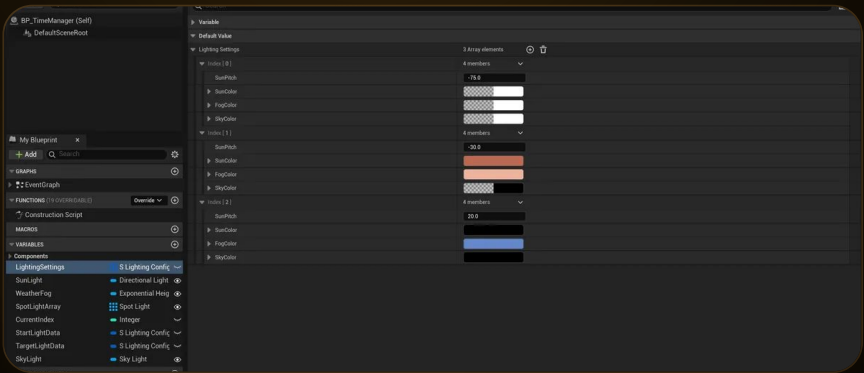


Fig 2. Fixed lighting parameters were stored as reusable states so each experimental condition could be reproduced consistently.

Experimental design

Nine participants completed all three conditions. A 3 by 3 Latin square balanced the order of Day, Sunset and Night to reduce learning, order and fatigue effects.

Participant measures

After each condition, participants rated visibility, search efficiency, attention, exploration, atmosphere and immersion on a five point Likert scale. Gameplay was also recorded for later observation.

Behaviour and analysis

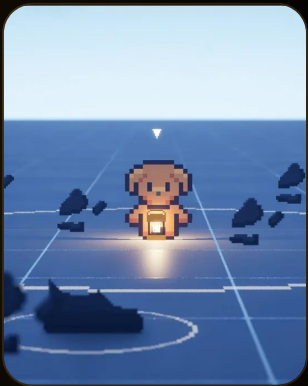
UE5 Blueprints recorded completion time and lantern duration. Lantern uses, fragments found through scanning and backtracks were also documented during each trial. Because the sample was small, the final analysis focused on descriptive statistics and cross condition patterns.

Technical Implementation



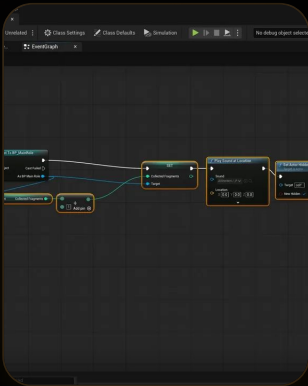
Engine and rendering

Unreal Engine 5 was selected for its real time Lumen global illumination and its support for 2D sprites inside a 3D scene. Directional Light, Sky Light and Volumetric Fog were tuned for each condition. Bloom and colour grading supported the visual atmosphere. Pixel assets were arranged across background, midground and foreground layers to create a controlled 2.5D space.



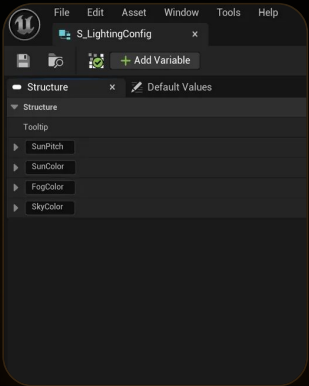
Lantern and scanning

The lantern combines local illumination with a spherical scanning volume. A Boolean state controls the Point Light and collision volume together. Overlap checks identify nearby fragments without adding visible prompts that could bias the experiment.



Fragment collection

An OnComponentBeginOverlap event validates the player actor. A successful pickup increments the collection count, plays audio feedback, hides the fragment and disables its collision. This keeps collection state consistent across trials.



Environment and data

A custom S_LightingConfig structure stores SunPitch, SunColor, FogColor and SkyColor. Timeline and Lerp nodes interpolate between fixed lighting states. The same system supports repeatable transitions and trial logging.

Prototype Development



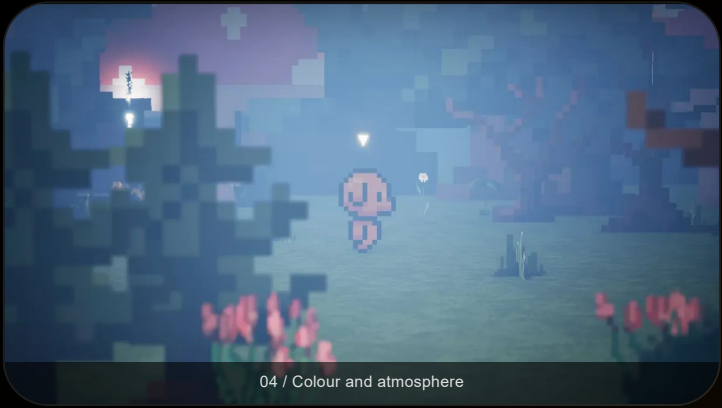
01 / Environment blockout



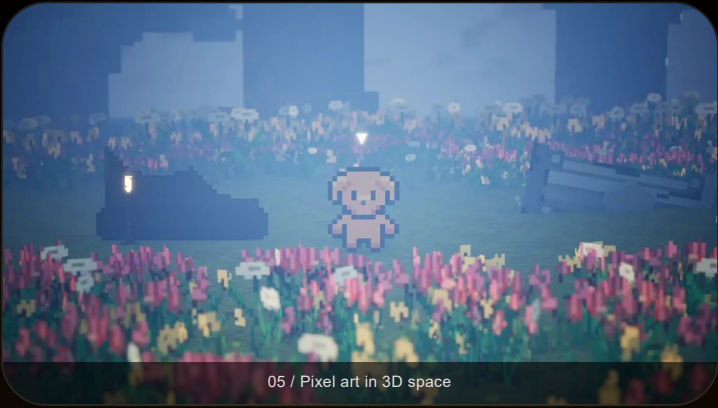
02 / Lumen lighting



03 / Low light test



04 / Colour and atmosphere



05 / Pixel art in 3D space

Results Findings

Across nine participants, the three lighting conditions produced distinct behavioural patterns. Lower illumination reduced perceived visibility and increased the need for careful search. It also increased lantern use, backtracking and task time. The strongest result was not a simple bright versus dark effect. Each condition changed how players searched.

OBJECTIVE BEHAVIOURAL MEASURES FROM NINE PARTICIPANTS

Measure	Day	Sunset	Night
Completion time (s)	106.01	96.26	124.41
Lantern-scanned fragments (mean)	0.33	1.44	3.00
Lantern uses (mean)	0.56	0.67	1.00
Lantern duration (s)	11.64	42.97	98.67
Backtracks (mean)	0.33	0.22	0.78

SUBJECTIVE RATINGS ON A FIVE POINT SCALE

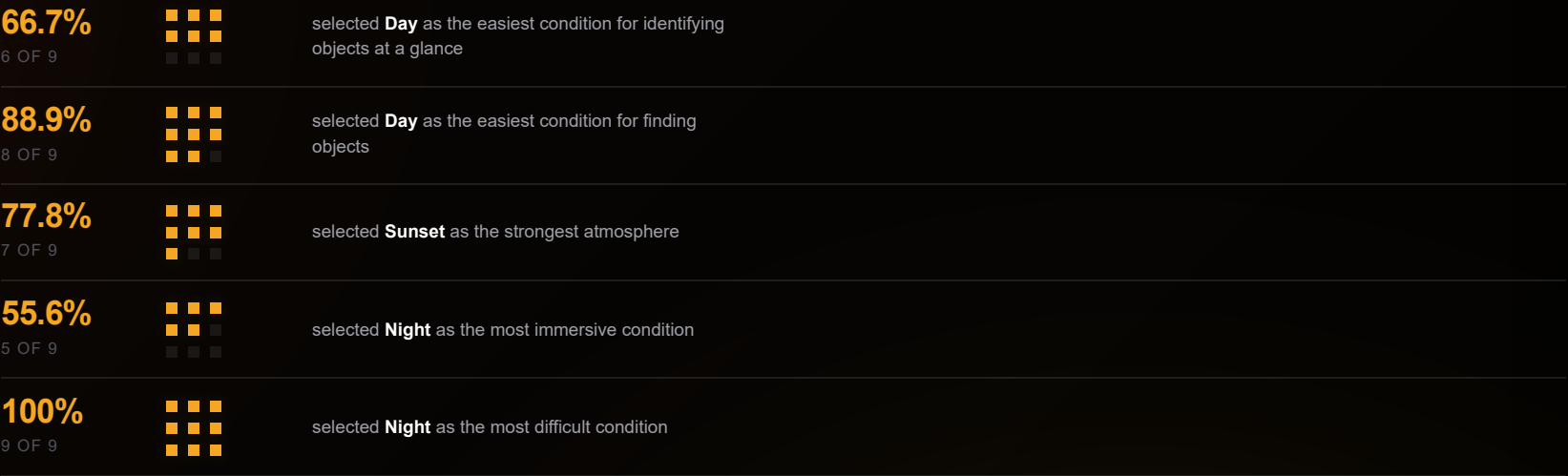
Visibility			Search Efficiency			Attention & Exploration		
Day		4.24	Day		3.08	Day		3.76
Sunset		3.78	Sunset		2.69	Sunset		3.72
Night		2.47	Night		1.75	Night		3.85

Means are calculated from the nine participants' five point ratings. Negatively worded search items were reverse coded so higher scores indicate more efficient searching.

Player Preferences

Participants did not equate visibility with experience quality. Day was consistently easier to read and search. Night was the most difficult condition, yet more than half of participants also found it the most immersive. Sunset produced the strongest atmosphere preference.

PARTICIPANT PREFERENCES FROM NINE PARTICIPANTS



PARTICIPANT VOICE

01

"The atmosphere at night feels the best as it feels like actually exploring a forest and guiding yourself through."

NIGHT / IMMERSION FEEDBACK

02

"Location memory of objects made it easier to find all required objects in day and sunset settings."

SEARCH BEHAVIOUR

What Darkness Changed

Difficulty and immersion did not move together.

Day produced the strongest visibility ratings, yet participants collected the fewest fragments. Night produced the weakest visibility ratings, yet every participant collected all three fragments. The difference came with a clear change in search strategy.

100%

of participants considered Night the most difficult condition.

55.6%

selected Night as the most immersive condition.

Night produced the lowest search efficiency rating at 1.75, compared with 3.08 in Day and 2.69 in Sunset. Participants also reported more scanning and slower searching. The qualitative responses suggest that reduced illumination changed how spatial information was acquired rather than simply making the scene darker.

Design Implications

Lighting can shape behaviour without changing the level itself. The findings suggest four practical principles for 2.5D pixel games.

Use light to guide behaviour

Reduced visibility encouraged slower movement, scanning and path revisiting. Bright conditions supported faster visual sampling but also produced more omissions.

Balance efficiency and immersion

Night created the strongest challenge and the highest immersion preference. Sunset offered the shortest completion time and the strongest atmosphere. Lighting can therefore be tuned to the intended experience.

Use diegetic tools as guidance

The lantern restored access to local information without adding an external interface prompt. Its use supported complete fragment collection in Night.

Treat lighting as a continuous parameter

Sunset combined the shortest completion time with the strongest atmosphere preference. This supports a graded approach to illumination rather than a simple bright or dark choice.

Reflection & Next Steps

Unreal Engine 5 was chosen because Lumen allowed real time changes in illumination while keeping the 2.5D scene intact. The lantern scan was implemented through an invisible collision volume. This avoided visual prompts that could have influenced participant behaviour.

The main limitation is the small sample of nine participants. The study therefore reports descriptive patterns rather than inferential claims. Future work could use a larger sample, eye tracking, more varied level geometry and finer lighting gradients. Separating the lantern's lighting and scanning functions would also help isolate their individual effects.

Participation was voluntary and anonymised. Informed consent and data protection procedures were followed under the University of York research ethics process. Generative AI was used only for technical assistance with Blueprint refinement and a small number of 2D textures. It was not used for data collection, recruitment, writing or analysis.

SHUHAN ZHANG / GAME DESIGN CASE STUDY IN DEVELOPMENT

LET'S SUPERMARKET

A low-poly management sim about turning everyday stock, pricing and checkout decisions into the growth of one small store.

MY ROLE

Game + interaction design

TEAM

Four collaborators

ENGINE

Unreal Engine 5

FOCUS

Management simulation



Checkout scene from the project page. The full frame is shown without cropping.

LET'S SUPERMARKET / 01 - 06

01 / CORE LOOP

STOCK TO SALE

Five small actions make one **business cycle.**

The first playable loop is being built around the decisions and labour of running the store, not just decorating it.

01	02	03	04	05
Order stock	Restock	Set prices	Open store	Checkout
Buy goods while wholesale costs are favourable.	Move boxes from the loading dock to the shop floor.	Choose a price customers are willing to pay.	Customers browse the floor and select products.	Scan items, take payment and make change.

Sales return as capital for more goods, staff and eventually a larger store.

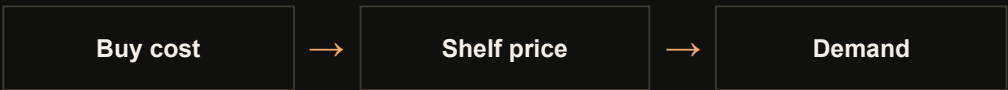
LET'S SUPERMARKET / 02 - 06

The shelf and the price tag both change the outcome.

SYSTEM A

Dynamic pricing

Wholesale costs fluctuate, creating moments to buy low. Each customer has an acceptable price range; demand drops when the store prices beyond it.



Design intent: profit depends on reading both supply and customer tolerance.

SYSTEM B

Store building

Modular zones can reinforce each other or create friction. The project page gives bakery and produce as a productive pairing, and a restroom beside the deli as a bad one.



A store is more than its inventory.

Staff behaviour and small narrative hooks give the management loop a social layer.

STAFF AI

Role, efficiency, morale

Recruited NPCs take a single assigned role. Their efficiency varies, and poor treatment lowers morale and productivity.

01 Recruit an NPC

02 Assign a role

03 Manage morale as well as output

STORY HOOKS

Milestones open quests

Revenue milestones are intended to trigger conversations with NPCs and a quest with a meaningful reward.

01 Reach a revenue threshold

02 Meet a selected NPC

03 Complete the quest for a payoff

Designing a playable slice.

MY CONTRIBUTION

Game and interaction design

I handle the game’s interaction and management design on a four-person team, alongside collaborators covering narrative and character / environment art.

Design	Core actions, store systems and player decisions
Narrative	Teammate role
Art	Teammate role

DEVELOPMENT STATE

First loop in progress

The team is using two starter goods to test pricing, shelving and basic building tools before expanding the demo.

BUILDING	Ordering, restocking, pricing, opening and checkout
PLANNED	More goods and buildable structures
PLANNED	Short story hook for the demo

A store day to prove the loop.

The first build tests the five-step loop with two starter goods.

The larger demo aims to add a handful of buildable structures, a couple of controllable characters, more sellable goods and a short story hook.

REPOSITORY

The project link remains clickable in the PDF, with a QR code for printed copies.



Explore the project
github.com/TaoMingxuan/Lets_Supermarket

Current: first playable loop in progress. More goods, buildables and story content are planned.

SHUHAN ZHANG / RESEARCH THROUGH DESIGN IN DEVELOPMENT

THRESHOLD

A physical conversation that asks when an artificial intelligence stops being itself.

FORMAT

Desktop-scale instrument

METHOD

Speculative design

FOCUS

Human-AI interaction

STATE

Build in progress



Proposed interaction schematic - not a photograph of a finished prototype.

What counts as an AI’s death?

Threshold turns an abstract question about identity into a short encounter with a visible, decaying object.

If memory, voice and relationship drift, when do we stop recognising the same entity?

The project uses a three-to-five-minute conversation. Six independent indicators decay quietly while the participant talks; neither side is told the values are falling. The final verdict is deliberately irreversible.

Context

Shutdowns of companion apps such as Replika and Soulmate form the project’s starting point.

Lens

Locke’s account of memory continuity and Parfit’s graduated view of identity frame the question.

Response

An embodied instrument asks for judgement rather than offering a companion relationship.

The change is felt before it is explained.

The participant observes a conversation losing continuity without a countdown or an announced failure state.

00	01	02	03	04
Idle	Connection	Free exchange	Decay	Verdict
The device waits, then speaks first.	A short baseline exchange, with every indicator near full.	The participant asks who, or what, they are talking to.	Indicators fall asynchronously; nothing announces it.	A dial and button demand one binding yes-or-no decision.

Design tension: the instrument measures six signals, but the human must decide what their combined loss means.

No single light says **alive**.

Each indicator turns a different form of continuity into a question the participant can notice during the exchange.

01

Memory

Can it still recall what we shared?

02

Identity

Has its voice drifted from who it was?

03

Relationship

Am I someone to it, or only a user?

04

Agency

Does it still reach out on its own?

05

Purpose

Does it know why it is here?

06

Survival instinct

Would it try to reach me if I left?

A physical loop for a **philosophical question**.

The planned stack links voice interaction, a decay engine and tangible outputs. The physical loop is still being built.

Hardware

Arduino Nano ESP32

WS2812 LED ring

E-ink / OLED display

Rotary encoder + button

Software

Python decay engine

OpenAI Responses API

Whisper + TTS voice loop

FastLED firmware

CURRENT STAGE

The hardware loop and decay engine are under construction. Prototype testing and reflection will follow.